

# LM73 2.7V, SOT, 11-to-14 Bit Digital Temperature Sensor with 2-Wire Interface

## FEATURES

- Single Address Pin Offers Choice of Three Selectable Addresses per Version for a Total of Six Possible Addresses.
- SMBus and I<sup>2</sup>C-Compatible Two-Wire Interface
- Supports 400 kHz Operation
- Shutdown Mode with One-Shot Feature Available for Very Low Average Power Consumption
- Programmable Digital Temperature Resolution from 11 Bits to 14 Bits.
- Fast Conversion Rate Ideal for Quick Power Up and Measuring Rapidly Changing Temperature
- Open-Drain ALERT Output Pin Goes Active When Temperature is Above a Programmed Temperature Limit
- Very Stable, Low-Noise Digital Output.
- UL Recognized Component

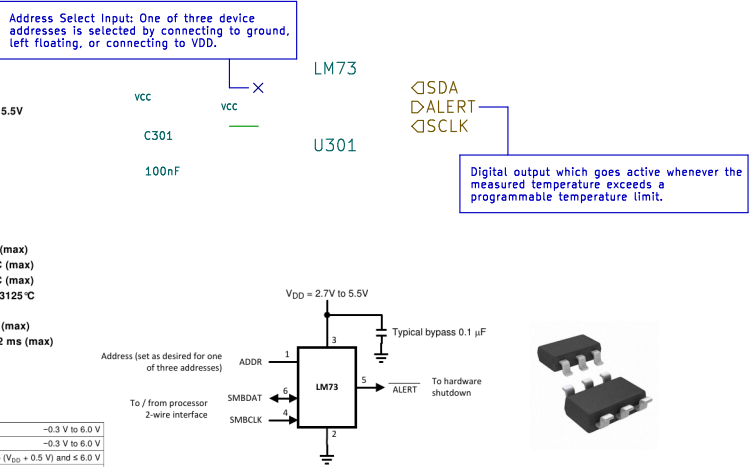
## KEY SPECIFICATIONS

- Supply Voltage: 2.7V to 5.5V
- Supply Current:
  - Operating:
    - 320  $\mu$ A (typ)
    - 495  $\mu$ A (max)
  - Shutdown:
    - 8  $\mu$ A (max)
    - 1.9  $\mu$ A (typ)
- Temperature Accuracy:
  - -10°C to 80°C:  $\pm$ 1.0°C (max)
  - -25°C to 115°C:  $\pm$ 1.5°C (max)
  - -40°C to 150°C:  $\pm$ 2.0°C (max)
- Resolution: 0.25°C to 0.03125°C
- Conversion Time:
  - 11-bit (0.25°C): 14 ms (max)
  - 14-bit (0.03125°C): 112 ms (max)

## ABSOLUTE MAXIMUM RATINGS<sup>(1)</sup>

|                                         |                                                      |
|-----------------------------------------|------------------------------------------------------|
| Supply Voltage                          | -0.3 V to 6.0 V                                      |
| Voltage at SMBCLK and SMBDAT pins       | -0.3 V to 6.0 V                                      |
| Voltage at All Other Pins               | -0.3 V to (V <sub>DD</sub> + 0.5 V) and $\leq$ 6.0 V |
| Input Current at Any Pin <sup>(2)</sup> | $\pm$ 5 mA                                           |
| Storage Temperature                     | -65°C to +150°C                                      |
| ESD Susceptibility <sup>(3)</sup>       | Human Body Model<br>Machine Model                    |
|                                         | 2000 V<br>200 V                                      |

Soldering process must comply with Texas Instruments' Reflow Temperature Profile specifications. Refer to [www.ti.com/packaging](http://www.ti.com/packaging).<sup>(4)</sup>



Mrk Industries

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Title: I<sup>2</sup>C digital temperature sensor

Size: A4

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